Evidence Based Practice 6: Outcome Measures in Clinical Practice

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Learning Outcomes

At the end of this session, students should be able to:

- Explain the use of clinical measurements in contemporary Evidence Based Practice (EBP)
- Discuss what is meant by psychometric properties of outcome measures
- Identify the types of outcomes and measures used in clinical physiotherapy and how to select the most appropriate OM



Assessment, measurement and evaluation

Assessment

- the action or an instance of making a judgment about something

- **Clinical assessment in healthcare**
  - an evaluation during which “information is collected to identify patient's needs and formulate a treatment plan”
  - evaluation of the patient using selected skills of history-taking; physical examination, laboratory, imaging, and social evaluation ...

Measurement

- ‘Measurement consists of rules for assigning numbers to observable attributes to represent quantities of the attributes’

  - Attributes – dimensions, properties, characteristics or behaviours -e.g muscle size, tone, sensation etc
  - Observability – generally attributes must be observable to be measured (objective measures)
  - Non-observable attributes

Evaluation

- **Evaluation**
  - the process of judging or calculating the quality, importance, amount, or value of something

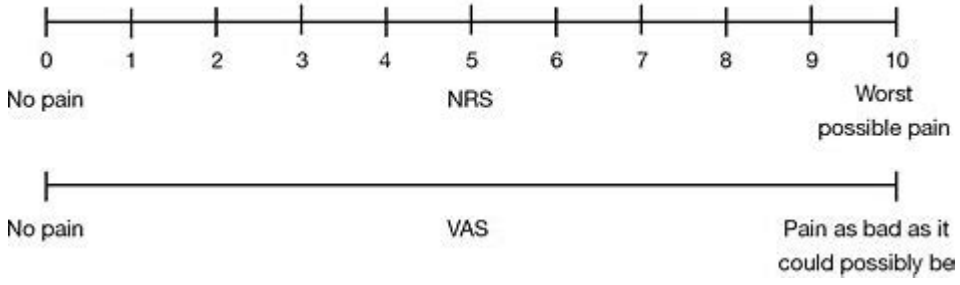
- **Medical/Physiotherapy Evaluation**
  - A medical evaluation is a comprehensive assessment of a patient's overall medical history and current condition for the purpose of identifying health problems and planning management

  - Patient evaluation is made through medical history, physical examination, routine laboratory tests, and other diagnostic procedures.

Sprained ankle



Grade	Muscle Activity
0	No contraction
1	Flicker / trace contraction
2	Active movement with gravity eliminated through full range
3	Active movement against gravity through full range
4	Active movement against gravity and resistance through full range
5	Normal power through full range

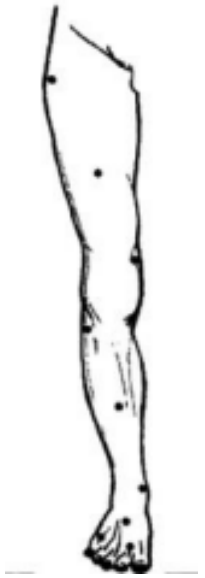




- ASSESSMENT
  - Subjective – **pain, function loss**
  - Objective – **Range of movement (ROM), muscle strength**
- CLINICAL MEASUREMENT to quantify signs, symptoms, functions and behaviours
  - **Goniometer –** degrees of movement available
  - **MRC Oxford Scale** – muscle strength
  - **Visual Analogue Scale (VAS) -** pain
- EVALUATION of patient status

Stroke

- ASSESSMENT
  - Subjective – sensation, mood, loss of function
  - Objective – muscle strength, functional ability, cognitive ability
- CLINICAL MEASUREMENT to quantify signs, symptoms, functions, behaviours
  - Barthel Index – functional independence
  - Nottingham Sensory Assessment – sensation
  - Beck Depression Inventory (BDI) – mood
- EVALUATION of patient status





1.		
	0	I do not feel sad.
	1	I feel sad
	2	I am sad all the time and I can't snap out of it.
	3	I am so sad and unhappy that I can't stand it.
2.		
	0	I am not particularly discouraged about the future.
	1	I feel discouraged about the future.
	2	I feel I have nothing to look forward to.
	3	I feel the future is hopeless and that things cannot improve.
3.		
	0	I do not feel like a failure.
	1	I feel I have failed more than the average person.
	2	As I look back on my life, all I can see is a lot of failures.
	3	I feel I am a complete failure as a person.

Why should you use clinical measurements?

To obtain baseline information

- Presence or absence of an attribute / Severity / Impact

For setting goals for treatment or management

- Think ICF

To evaluate client's ability and capacity to progress

- In exercise programme / prognosis / independence

To evaluate efficacy of treatment/management intervention

- Improvement / Deterioration / No change

Research to build evidence for Evidence Based Practice

What is an outcome measure?



Outcome measures generate scores that are intended to quantify a patient's performance or health status based on standardized evaluation protocols or close-ended questions *(Jette et al. 2009)*

What can be measured or quantified?

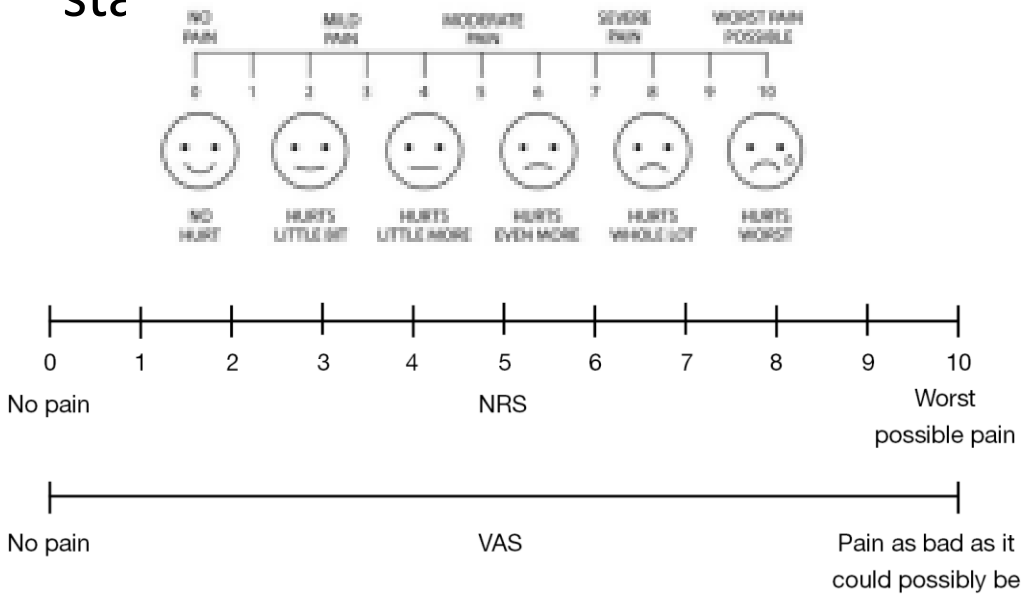
- **Objective measures** – measuring something that can be observed
  - E.g., height, weight, range of movement (ROM), muscle strength, behaviour







- **Subjective measures** – measuring something that the patient/individual reports to you
  - e.g. pain, pins & needles (paresthesia), mood, psychological sta



Types of Outcome measures

- **Performance based measures**
  - Patient must be observed performing tasks
  - Measured quantitatively e.g., time taken
  - Measured qualitatively e.g., performed independently, required some assistance, unable to perform
  - Measures capacity not habitual performance (i.e. - what patient can do) right now (at a particular point in time)
- **Observer reported measures**
  - Measures completed by another person e.g., parent, carer, partner
  - May indicate habitual performance better (i.e. - what patient does do)
- **Self-report or patient reported outcomes (PROs)**
  - Subjective experience is quantified
  - Measures habitual performance (i.e. - what patient does do)
- **Clinician reported measures**
  - Measures completed by a health care professional
- **Disease specific or generic**

Psychometric properties of Outcome measures

RELIABILITY – stability over time if nothing has changed

VALIDITY – a tool measures what it intends to measure

RESPONSIVENESS – can detect changes in conditions or differentiate between groups

STANDARDISED outcome measures – standardise the assessment/measurement process to enhance the validity, reliability and hence the responsiveness

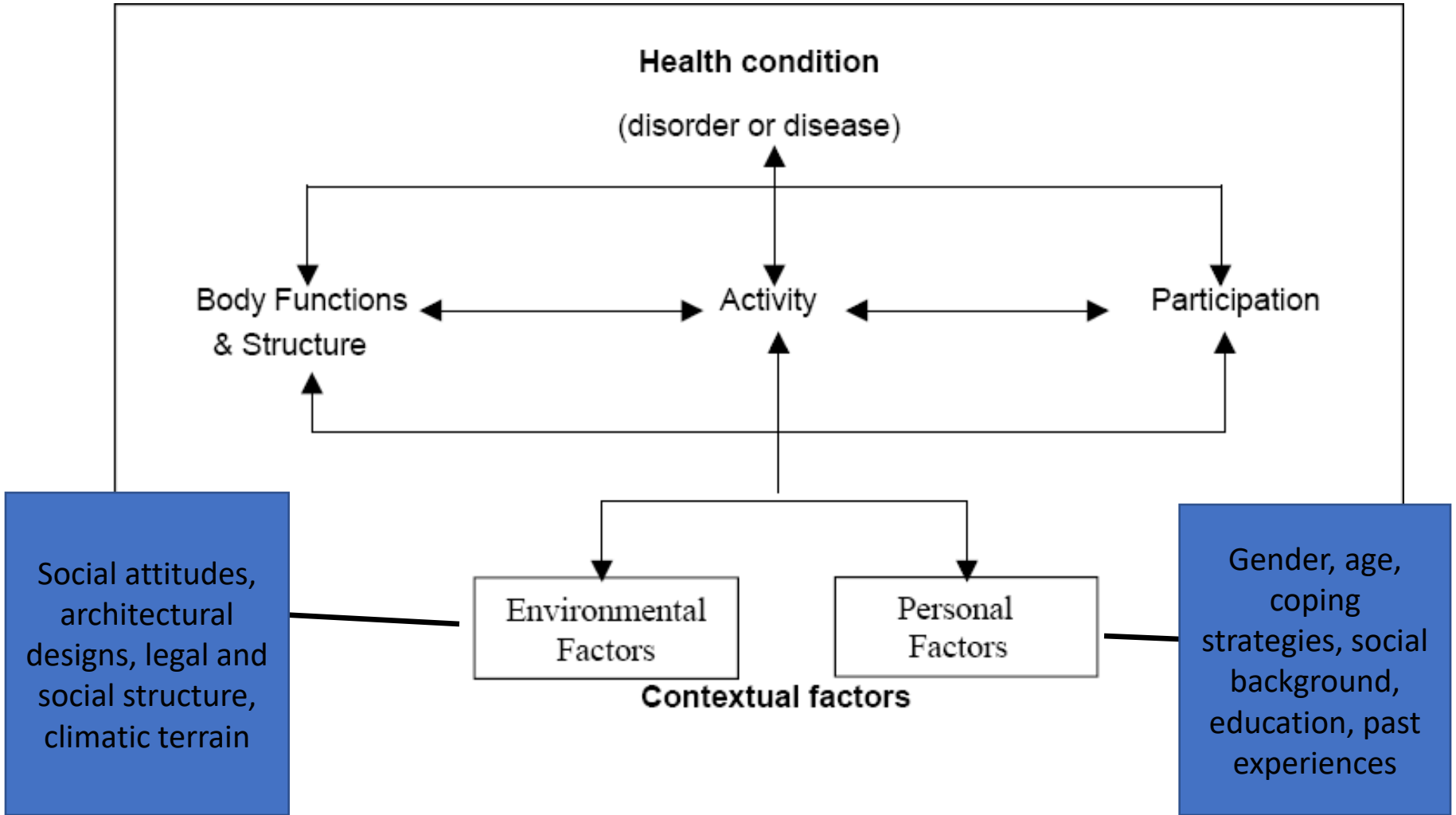
Selecting Outcome measures – factors to consider

1. What to Measure	Body Function/structure, activity , participation
2. Purpose of measure	Discriminative, predictive or evaluative
3. Type of measure	Disease specific or generic
	Self-reported or performance-based
4. Patient and clinic factors	Patient recovery and ability
	Patient goals
	Clinic requirements
5. Psychometric properties	Reliability, validity, diagnostic accuracy, responsiveness
6. Feasibility	Time, space and equipment
	Training required
	Cost
	Respondent burden, culture and language

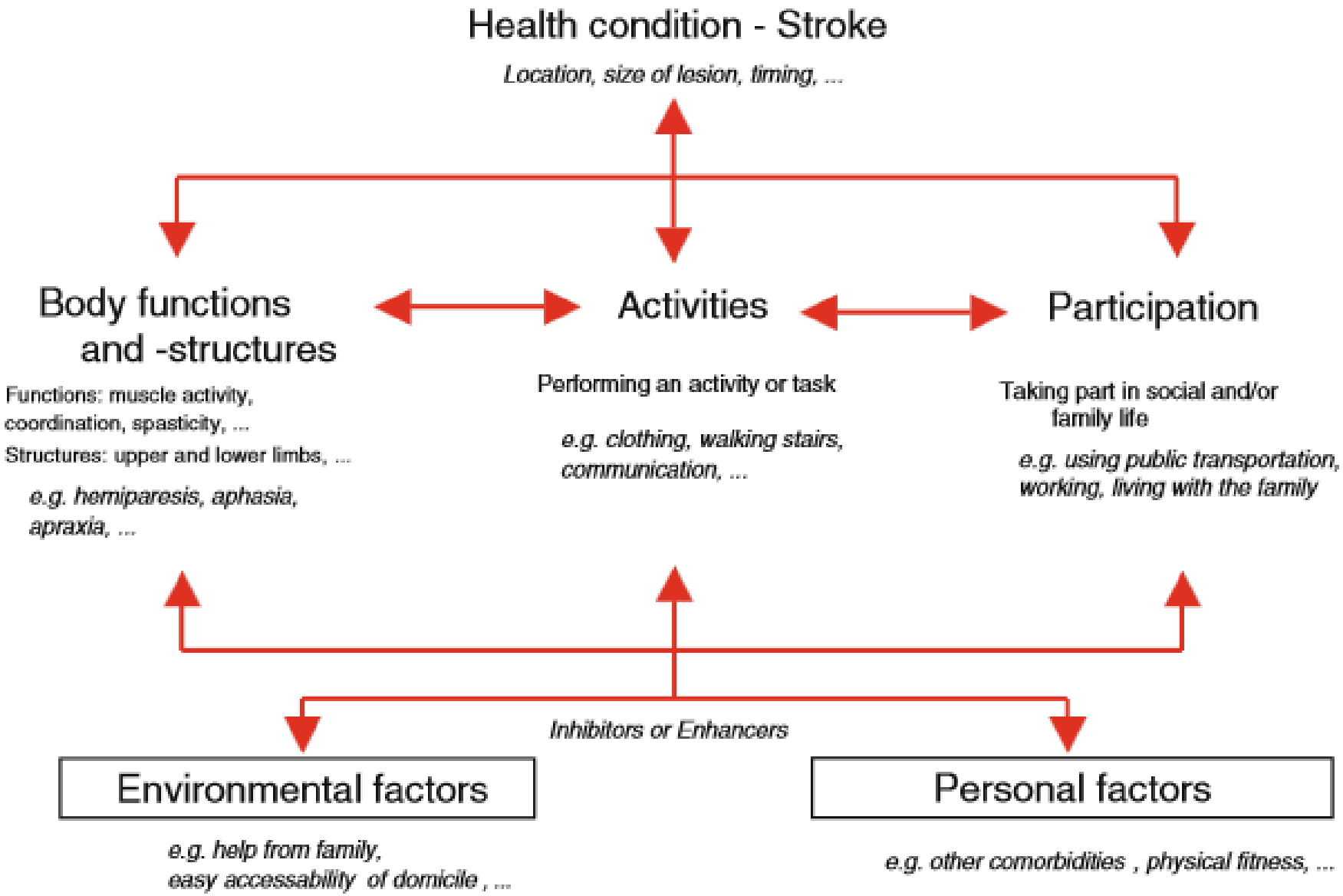
What to measure

- ICF is used to decide what to measure
- You should consider using outcome measures across all levels of the ICF - to gain a broad understanding of the patient's health condition.
- During the history taking, you identify potential constructs that may be important to measure
  - Identification of relevant categories of tests and measures
    - Specific parameters to measure – and subsequently the appropriate measure.

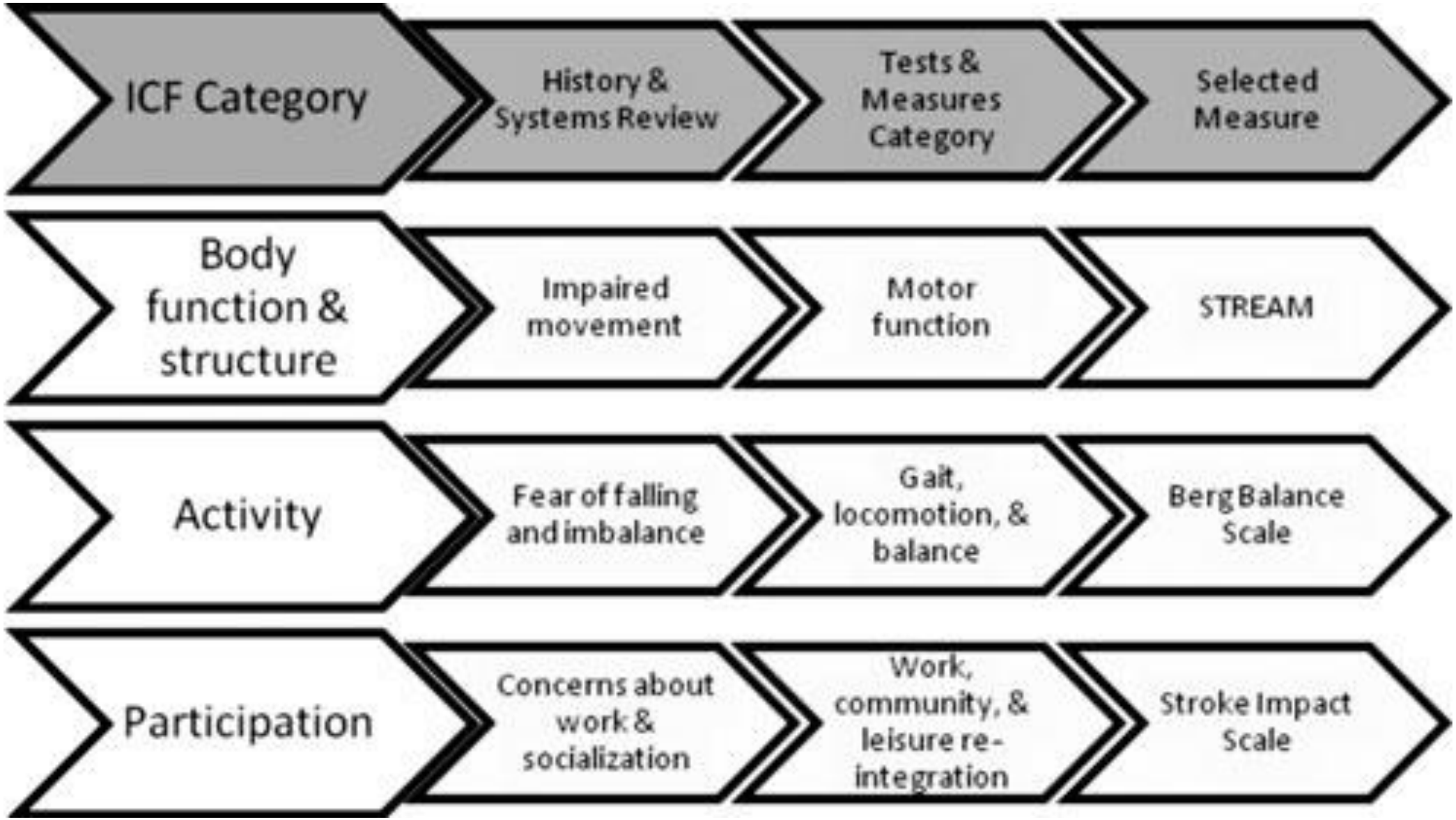
International Classification of Functioning, Disability and Health (ICF, 2001)



ICF –with Stroke as an example



J Neurol (2012) 259:817–832


ICF Category
History & Systems Review
Tests & Measures Category
Selected Measure
Body function & structure
Impaired movement
Motor function
STREAM
Activity
Fear of falling and imbalance
Gait, locomotion, & balance
Berg Balance Scale
Participation
Concerns about work & socialization
Work, community, & leisure re-integration
Stroke Impact Scale



Lt MCA infarct 2 weeks after incident

Poor sitting balance, inability to stand, require assistance of two for transfers

52-year-old, married father of 2, lives in a council flat on 6th floor, loves to go to pub, works

Pain, muscle weakness, spasticity, loss of inter-joint co-ordination, shoulder subluxation.

VAS, Oxford muscle grading, MAS, etc

Clinical decision-making framework – (Potter et al. 2011)

Selecting Outcome measures – factors to consider

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Purpose of the measure

- Generally, OMs are used for one of 3 reasons:
  - **Discriminant**
    - to distinguish between individuals or groups based on a particular characteristic
  - **Predictive**
    - to forecast future status and can be used to identify risk, determine a prognosis, and plan discharge e.g., Berg Balance scale
  - **Evaluative**
    - Determine the effectiveness of an intervention by evaluating change over time. e.g., BBS, gait speed

Type of measure

- Outcome measures can also be classified according to focus (generic or disease-specific) and mode of administration

Generic vs Specific

- Generic OMs – used across all patient populations regardless of specific health condition e.g., Barthel Index, SF-36, Functional Independence measure (FIM)
- Specific e.g., Parkinson Disease Questionnaire -39 (PDQ -39)

Performance-based vs self-reported

- Performance – patient performance in a specific environment – Berg Balance Scale
- Self-report OMs – patients' opinions on impact of condition – Activities Specific Balance Confidence scale

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Patient & Clinic Factor

Patient Factors

- Patient's goals and the planned intervention
- Select across ICF important – meaningfulness, intervention specificity
- Stage of recovery
  - Early – body structure/function, activity
  - Later – measures of participation
- Patient's capability – not too difficult – ceiling and flooring effects

Clinic Factors

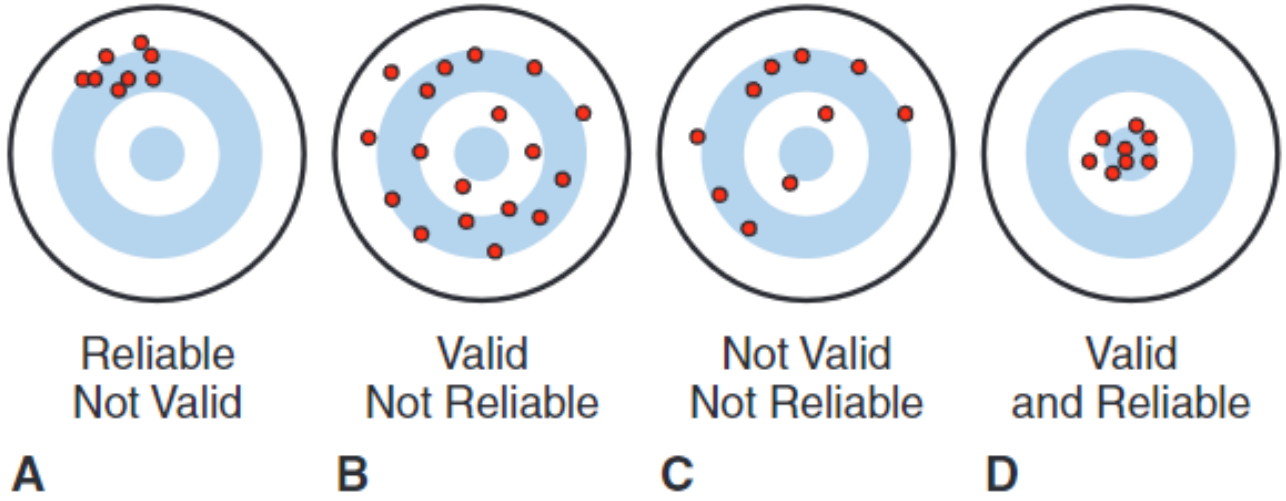
- Some settings require that certain OMs be administered with all patients e.g., FIM

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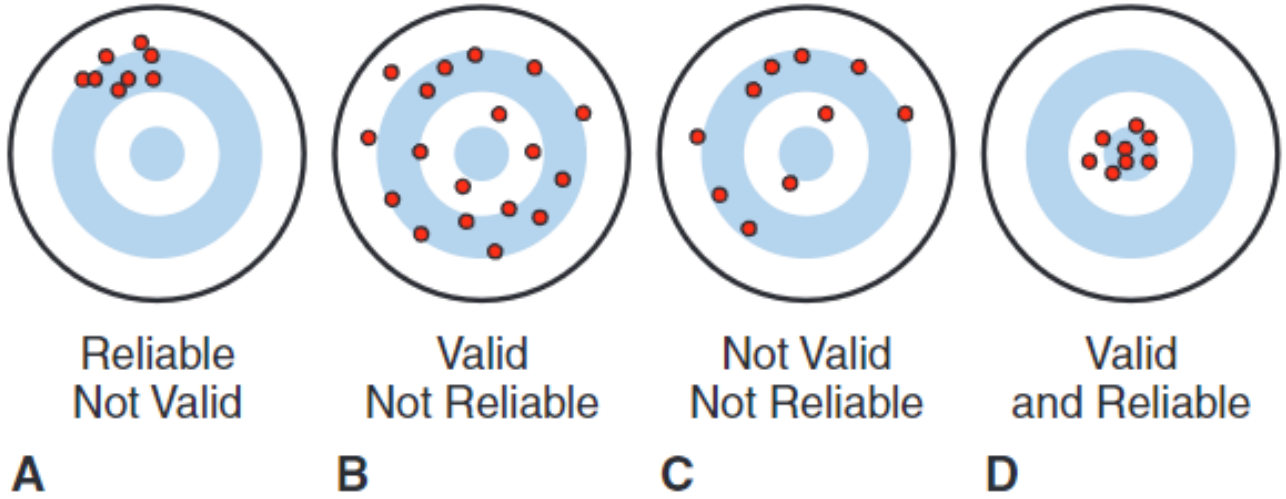
Psychometric Factors

- Reliability and Validity
- Diagnostic accuracy and responsiveness
- Important Psychometric properties
  - Minimum Detectable change (MDC)
  - Minimum Clinically Important Difference (MDIC)



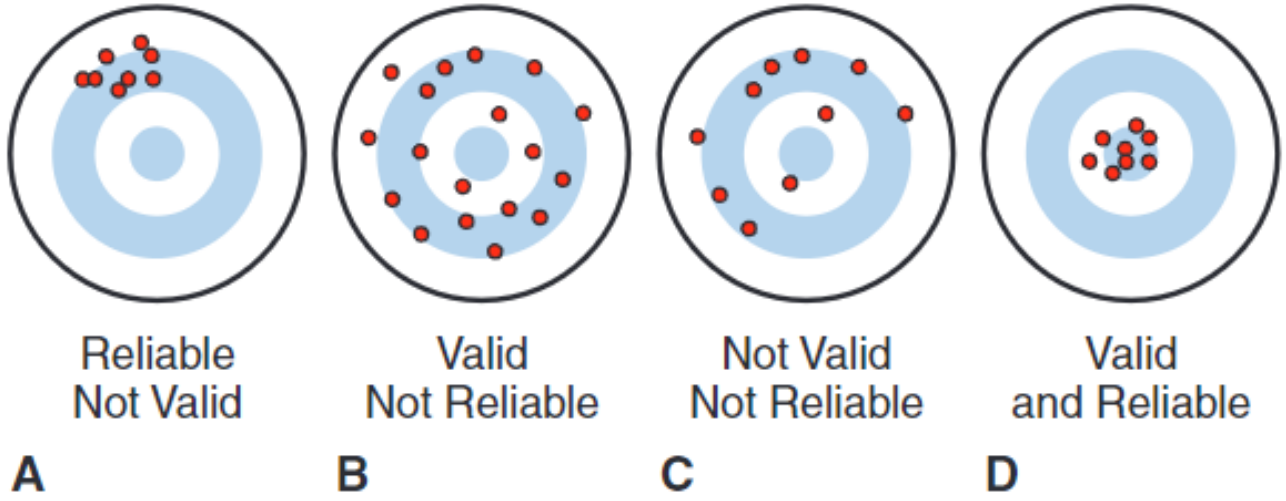
Psychometric Factors

- Reliability and Validity
- **Diagnostic accuracy and responsiveness**
  - Sensitivity: the ability of a test to correctly identify patients with a disease
  - Specificity: the ability of a test to correctly identify people without the disease.
- Important Psychometric properties
  - Minimum Detectable change (MDC)
  - Minimum Clinically Important Difference (MDIC)



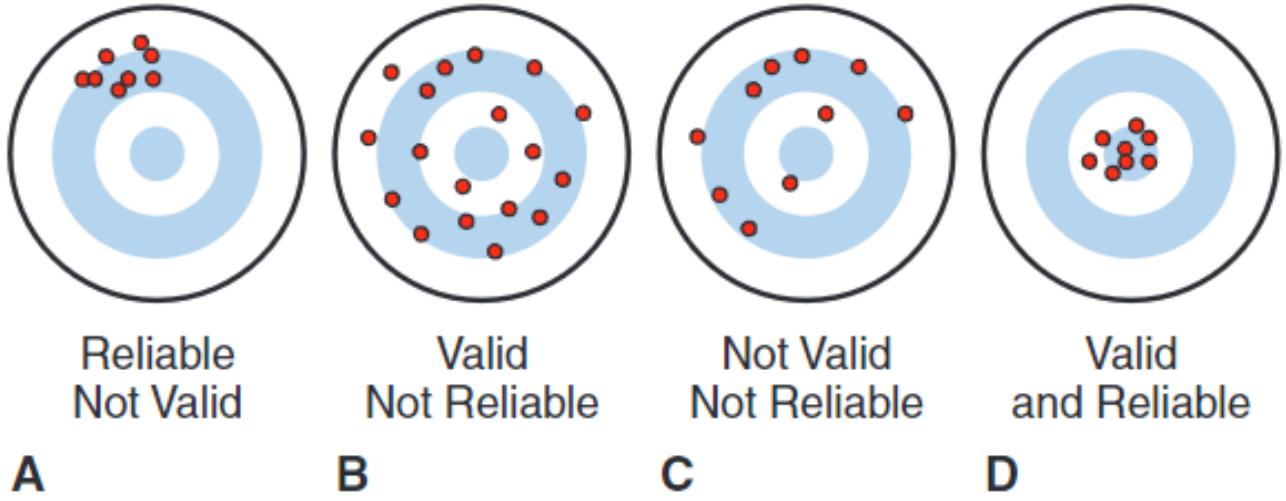
Psychometric Factors

- Reliability and Validity
- Diagnostic accuracy and responsiveness
- Important Psychometric properties
  - **<u>Minimum Detectable change (MDC):</u>** is the minimal amount of change on the OM that is necessary to exceed error and variability in that measure.
  - Minimum Clinically Important Difference (MDIC)



Psychometric Factors

- Reliability and Validity
- Diagnostic accuracy and responsiveness
- Important Psychometric properties
  - Minimum Detectable change (MDC
  - **<u>Minimum Clinically Important Difference (MCID):</u>** The MCID is defined as the smallest difference in a measure that is considered worthwhile or important

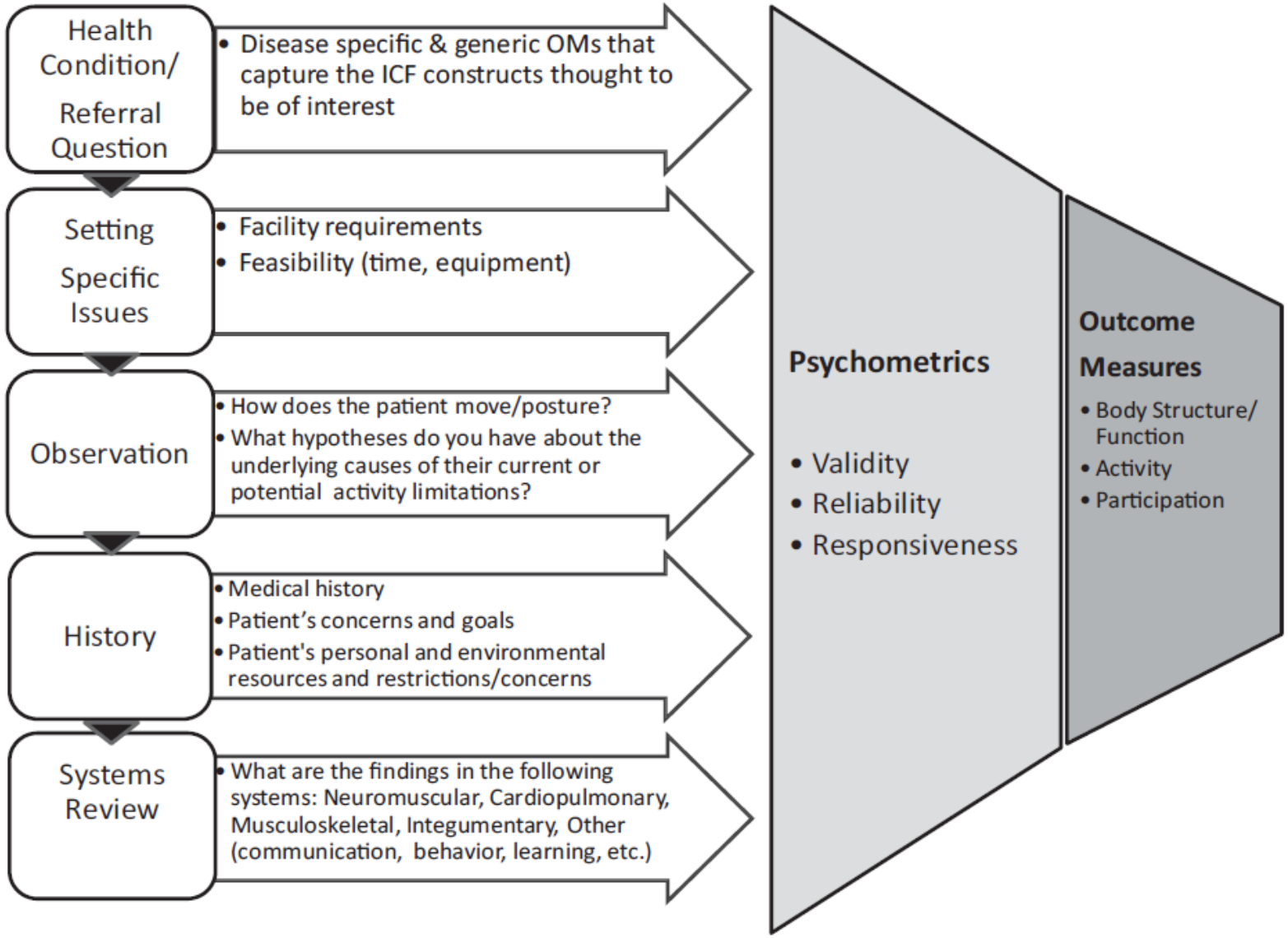


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Feasibility

- Time e.g BESTest vs mini-BESTest
- Space – e.g. 6 minutes' walk test
- Equipment requirement e.g. kinematics & kinetic evaluations
- Training
- Culture and language
- Cost of use – proprietary issues



Selecting outcome measures

Body Structure/Impairment outcome measures

- https://www.sralab.org/rehabilitation-measures

IMPAIRMENT	OUTCOME MEASURES
Motor function	1. Fugl-Meyer Motor Assessment 2. Motor Assessment Scale 3. Stroke Rehabilitation Assessment of movement 4. Motoricity Index
Spasticity	1. Modified Ashworth Scale 2. Tardieu Scale
Sensation	1. Nottingham Sensory Assessment
Neglect	1. Behavioural Inattention Test
Cognition	1. Mini-Mental State Assessment 2. Montreal Cognitive Assessment

Activity Outcome Measures

Impairment	Outcome measures	Impairment	Outcome measures
Mobility	1. 6-meter walk test 2. 6 minutes' walk test 3. Modified Rivermead Mobility Index 4. Functional Ambulation Categories 5. Dynamic Gait Index 6. Functional Gait Index	Balance	1. Berg Balance Scale 2. Activity Balance Confidence Scale 3. BESTest 4. Mini-BESTest 5. Functional Arm Reach
Arm Function	1. Nine Hole Peg Test 2. Action Research Arm Test 3. Frenchay Arm Test	Global Disability/ADL measures	• Barthel Index • Functional Assessment Measure (FIM+FAM) • Health Assessment Questionnaire

Participation Outcome measures

Participation domain	Outcome measures
Participation	1. London Handicap Scale 2. Medical Outcomes Study Short Form 36 (SF-36) 3. Nottingham Health Profile 4. Stroke Impact Scale 5. Canadian Occupational Performance Measure (COPM)
Arm Specific Participation	1. Motor Activity Log

Some disease-specific Outcome measures

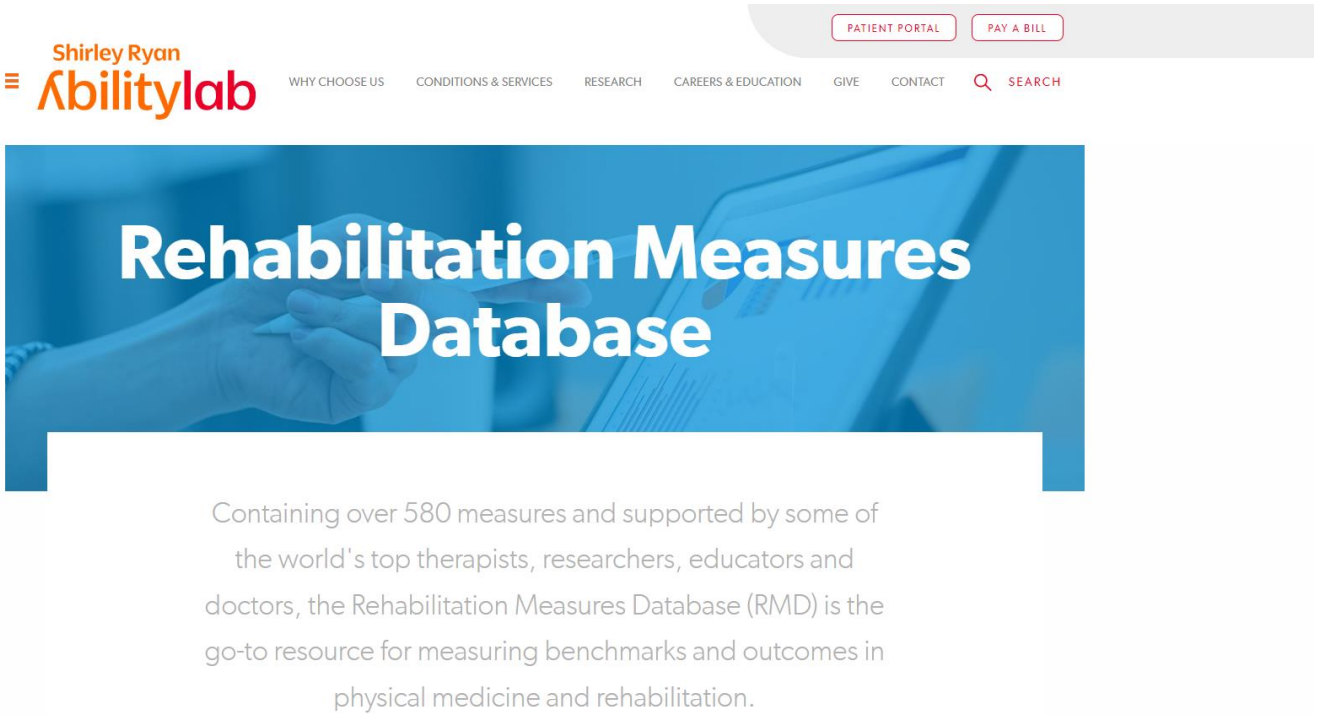
- Multiple Sclerosis
- Parkinson Disease
- Ataxia
- Osteoarthritis
- Pain
- Patient satisfaction
- Physical activities

References

1. Jette DU, Halbert J, Iverson C, Miceli E, Shah P. Use of standardized outcome measures in physical therapist practice: perceptions and applications. Phys Ther. 2009;89:125-135.
2. Van Peppen RP, Maissan FJF, Van Genderen FR, Van Dolder R, Van Meeteren NLU. Outcome measures in physiotherapy management of patients with stroke: a survey into self-reported use, and barriers to and facilitators for use. Physiother Res Int. 2008;13:255-270.
3. https://www.sralab.org/rehabilitation-measures
4. www.ebrsr.com

Outcome measure Workshop

Outcome measures



https://www.sralab.org/rehabilitation-measures

Body Structure/Impairment outcome measures

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Evaluating an outcome measure for use in clinical practice

Domain	
Evaluation	Questions to ask
Aim of the outcome measure • Is there information about why and how the measure was developed • How is it scored	What does the instrument plan to measure and wa it originally designed for that purpose?
Content	Describe the Outcome measure
User centredness • Whose perspective does it capture? • What domains are covered? • What is the method of administration?	• Does the measure capture outcomes that are required by various users? • Are the outcomes meaningful to the patient/client, the healthcare professional, a carer, the service provider or commissioner? • Is the measure an observation of performance, a measure of opinions/attitudes or a self-reported measure of impact or performance • How does the scoring relate to the raw opinion or performance? Are there any distortions or biases present in the way the information is captured?

Evaluating an outcome measure for use in clinical practice

Domain	
Content	Questions to ask
Measurement properties • What evidence is there about reliability with this population? • What evidence is there about its validity with this population? • Is there information about how responsive to meaningful clinical change it is?	Validity, reliability, measuring change
Feasibility	
Is there training required? Lots of equipment? Cost?	• How feasible is this OM to use in your clinical practice? • Will the OM yield good quality data? • Is the data easy to analyse and will reports be meaningful to both patients/clients and other interested parties? • Is there a cost associated with its use?
Utility How acceptable is it for you, for your patients?	• Will gathering this information be valuable to the patient/client? • Will it be used in clinical decision making? • Does it provide unique information not available elsewhere? • Will it capture the impact of a service